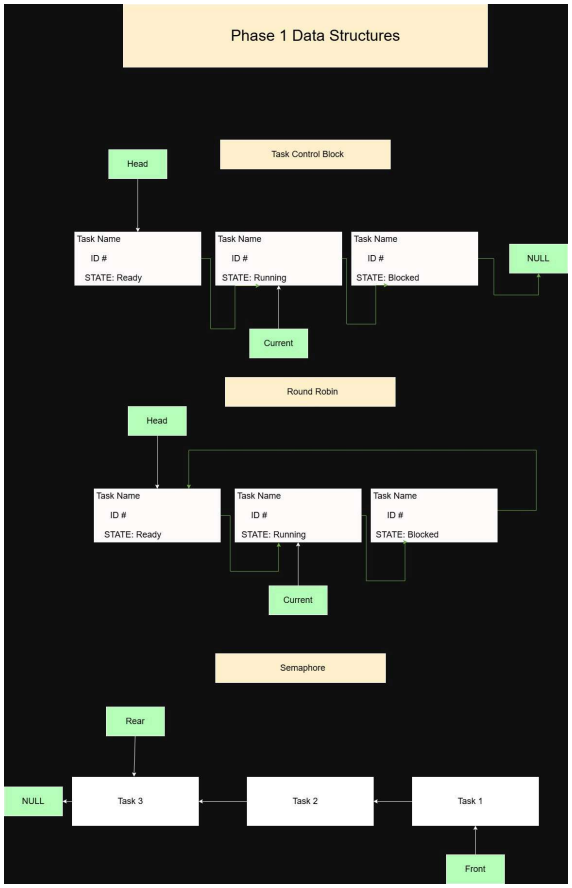
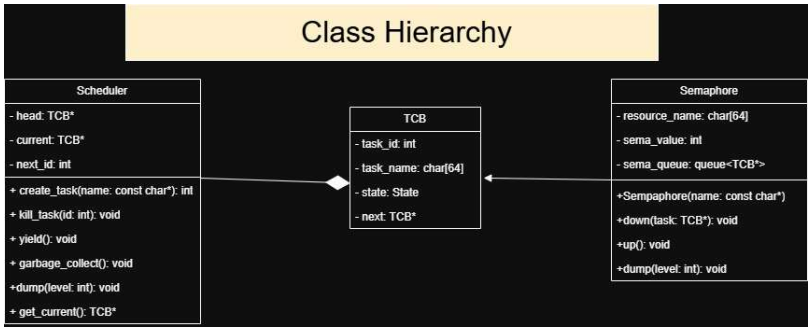
Design Diagrams:

The data structure is the tasks organized by the TCB, using a single linked list, where each TCB contains a ‘next’ pointer, pointing to the next task in the list. This allows the scheduler to move, add, and remove tasks. The scheduler has a pointer to the head of the list and one pointer to the current executing task, enabling round-robin scheduling. The semaphore uses a queue data structure to manage blocked tasks. When a task is blocked due to the unavailability of resources, it is placed inside the queue with FIFO order.

The scheduler class controls the collection of TCB objects. This is a composition relationship between the scheduler and TCB. The TCB depends on the scheduler as the scheduler creates, links, and removes TCB objects. Semaphore has an association relationship with TCB. The semaphore only controls access to shared resources with blocking and unblocking tasks.